Solo project: Gabriel Dautovi

AI agents now open large numbers of pull requests, but merge rates say little about whether merged code lasts. Code that is quickly rewritten carries hidden maintenance costs. The AIDev dataset (Li et al., 2026) documents agent PRs at scale, yet mostly measures acceptance and review outcomes. Human code lifetimes are well characterized (Spinellis et al., 2021), and Rahman and Shihab (2026) recently found that agent generated lines survive longer than human lines overall (hazard ratio 0.84), but report that variation between agents exceeded the agent–human gap, leaving it unclear which agents and task types drive this result. This study replicates and extends their analysis on AIDev's curated subset of popular repositories. For merged agent PRs, I will clone each repository, trace surviving lines with “git blame”, and compare them to matched human lines from the same repositories using Kaplan–Meier curves and Cox models stratified by agent, task type, and PR size. I will also classify why agent lines are modified (bug fix, refactor, revert). I expect agent code to last longer than human code overall, as Rahman and Shihab found, but not for every agent or every kind of change. Specifically, I expect at least one agent, or feature adding PRs in general, to have code that is modified or deleted sooner than comparable human code. If agent code lasts at least as long as human code for every agent and task type, this hypothesis is wrong.

- Li, H., Zhang, H., & Hassan, A. E. (2026). AIDev: Studying AI Coding Agents on GitHub. arXiv:2602.09185.
- Rahman, M., & Shihab, E. (2026). Will It Survive? Deciphering the Fate of AI-Generated Code in Open Source. arXiv:2601.16809.
- Spinellis, D., Louridas, P., & Kechagia, M. (2021). Software evolution: the lifetime of fine-grained elements. *PeerJ Computer Science*, 7, e372. https://doi.org/10.7717/peerj-cs.372